

Bruno Ferreira

Applied AI Engineer | Python · LLMs · RAG · Document automation

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Applied AI Engineer with 4+ years across AI research, university teaching and client projects. I design and build NLP, LLM and document-automation solutions using Python, RAG and full-stack engineering. My client work spans LegalTech, technical-specification extraction and quality-document validation, with direct responsibility for solution design, implementation and evaluation within each project scope.

Technical skills

AI engineering: LLMs, RAG, embeddings, NLP, OCR, information extraction, evaluation

Software: Python, FastAPI, TypeScript, React, Next.js, SQL, PostgreSQL, Docker, Git

Delivery: Process discovery, requirements, API integration, testing, documentation, training

Professional experience

Applied AI Engineer and Consultant

Independent client work · Portugal and remote | Jun 2025–present

- Own technical delivery within each client project scope: workflow analysis, solution architecture, implementation, testing and handover documentation.
- Advise on AI use cases and acceptance criteria; design and deliver practical training for client and developer teams.

Selected client projects

LegalTech platform for US visa workflows

2025–present · Components in development

- Designed and implemented intake, triage, dynamic questionnaires and document-management components in a multi-tenant application.
- Combined OCR and evidence-guided LLM analysis to prepare reports for legal review against USCIS criteria.

Python, FastAPI, Next.js, PostgreSQL, Redis, Alembic, Docker, OCR, LLMs

HVAC specification extraction

Jan–Mar 2026 · Technical project

- Built a Python pipeline combining embedding-based RAG, table mapping, regular expressions and vision/LLMs to extract specifications from Portuguese PDFs.
- Converted PDF specifications into structured Excel and JSON for engineering review, retaining source-page evidence and evaluating extraction against manual reference data.

Python, RAG, embeddings, PDF/table extraction, vision, LLMs, Excel, JSON

Quality Copilot for packaging and technical documents

2026 · Private pilot

- Built modules for dimensional checks, content validation, correction verification and technical-sheet preparation.
- Defined the evidence required at each stage and generated reports for human review.

Python, OCR, Claude Code skills, document automation

Teaching and research

Invited Lecturer

University of Coimbra | Feb 2024–present

- Teach databases, AI, Java, C, natural language processing and knowledge representation.
- Prepare learning materials, assess student projects and supervise MSc dissertations.

Researcher

CISUC · Centre for Responsible AI | Feb 2022–Jun 2025

- Researched NLP, information extraction, question answering and the evaluation of knowledge encoded in language models.
- Published work on language-model evaluation and document information extraction, alongside research in explainable AI, causal inference and model compression.

Education

PhD in Informatics Engineering

University of Coimbra · 2022–present · In progress

MSc in Informatics Engineering — Intelligent Systems

University of Coimbra · 2022

BSc in Informatics Engineering

University of Coimbra · 2021

Additional work

Privacy-focused AI platform · Own product in development

Built a functional React/TypeScript PWA using local NER and ONNX for personal-data detection and anonymisation before external LLM workflows.

THE 17.3 evidence evaluation · Commissioned project, 2025

Designed and carried out a repeatable review of public university evidence across 17 THE Impact Rankings indicators.

Selected publications

[Towards Automated Evaluation of Knowledge Encoded in Large Language Models](#) · DLnLD @ LREC-COLING 2024

[BATS-PT: Assessing Portuguese Masked Language Models in Lexico-Semantic Analogy Solving and Relation Completion](#) · PROPOR 2024

[Leveraging Question Answering for Domain-Agnostic Information Extraction](#) · CIARP 2023

[Full publication list and project details](#)

Languages and availability

Portuguese: native · English: advanced · Spanish: good comprehension, basic speaking

Based in Pombal, Portugal. Open to Applied AI / AI Solutions roles and B2B engagements through my company, including defined projects and ongoing contracts.

Development tools: Claude Code, Codex, OpenCode and Pi, with my own code review and testing.